

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA101: Discrete Structures (New Course)

MCA111: Discrete Mathematical Structures (Old Course)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

$$\begin{array}{r} 23! \\ \hline (23-3)! \cdot 3! \end{array}$$

Answer TWO questions.

2×12=24

1. Give the definition of a group. Show that the group G is Abelian if and only if $(ab)^2 = a^2b^2$.
- 2(a) ✓ What is a proposition? Under what condition two or more propositions will be equivalent? Determine whether the compound propositions $p \leftrightarrow q$ and $(p \wedge q) \vee (\sim p \wedge \sim q)$ are equivalent propositions by constructing a truth table.
(b) ✓ Explain few applications of pigeonhole principle. A bag contains 10 red marbles, 10 white marbles and 10 blue marbles. What is the minimum number of marbles we have to choose randomly from the bag to ensure that we get 4 marbles of same colour.
- 3(a) ✓ The spring dance committee must have 3 freshmen and 5 sophomore members. If there are 23 eligible freshmen and 18 eligible sophomores, how many different committees are possible?
(b) ✓ Let $A = \{1, 2, 3, 4\}$. Suppose $R = \{(1, 1), (1, 2), (2, 3), (2, 4), (3, 4), (4, 1), (4, 2)\}$ and $S = \{(3, 1), (4, 4), (2, 3), (2, 4), (1, 1), (1, 4)\}$ be two relations on A .
(i) ✓ Find $R \circ S$. (ii) ✓ Find the transitive closure of R .

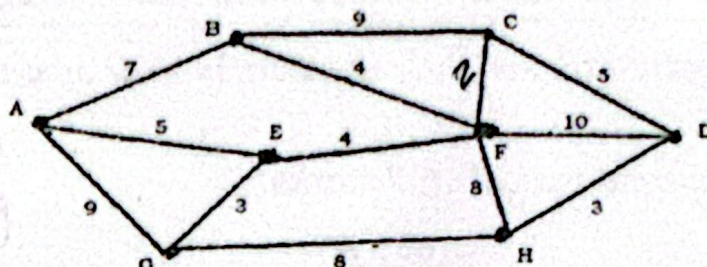
Contd. ...

(2)
Group B

Answer SEVEN questions.

7×8=56

4. What is a tree? Define minimal spanning tree for a graph. Use PRIM's algorithm to find a minimal spanning tree for the given graph. List the edges in the order in which they are chosen.



5. Give the definition of a permutation function on a finite set A.

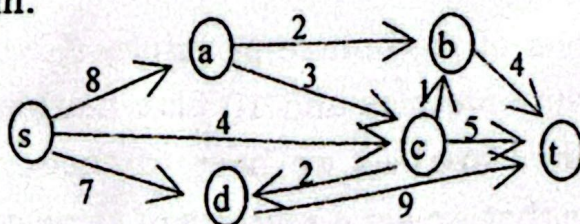
Let $p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{pmatrix}$

- (i) Write p as a product of disjoint cycles.

- (ii) Compute p^2 .

- (iii) Determine whether p is even or odd permutation.

6. What is a maximal flow in a transport network? Find the maximal flow for the network in the given figure using labeling algorithm.



7. Prove that the statement is true by using mathematical induction for all positive integers n ,

$$P(n) = 1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{(3^n - 1)}{2}$$

8. What is a Karnaugh map? Prepare the truth table for the Boolean expression

$$f(x, y, z) = (x' \wedge y \wedge z') \vee (x \wedge y' \wedge z') \vee (x \wedge y' \wedge z) \vee (x \wedge y \wedge z).$$

Construct the Karnaugh map to simplify the above Boolean expression.

Contd. ...

(3)

9. Give the definition of homomorphism of groups with example. Let G be the group of integers under the operation of addition and G' be the group of all even integers under the operation of addition.

Show that the function $f: G \rightarrow G'$ defined by $f(a) = 2a$ is an isomorphism.

10. What do you understand by a linear homogeneous recurrence relation with constant coefficients? Solve the recurrence relation for the given initial conditions.

$$b_n = 2b_{n-1} + 8b_{n-2} : b_0 = 4, b_1 = 10$$

11. Define Hamiltonian path and Hamiltonian circuit. Let the number of edges of a graph G be m . Then G has a Hamiltonian circuit if $m \geq \frac{1}{2}(n^2 - 3n + 6)$, Where n is the number of vertices.

12. Write short note on (Any TWO):

- (a) Finite state Automata
- (b) Error correcting codes
- (c) Methods of proof

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$$\frac{1}{2}(n^2 - 3n + 6) = \frac{1}{2}n^2 - 3n + 6 -$$